

SKILL - 18

Code:

```
GNU nano 7.2 skill18_job_control.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/types.h>
#include <sys/wait.h>

volatile sig_atomic_t running = 1;

void handle_signal(int sig)
{
    if (sig == SIGTERM || sig == SIGINT) {
        running = 0;
    }
}

int main()
{
    pid_t pid;
    int status;

    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);

    printf("OS Skill-18: Signal and Job Control\n");
    printf("Parent Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        signal(SIGINT, SIG_DFL);
    }
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ gcc skill18_job_control.c -o skill18_job_control
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ ./skill18_job_control
OS Skill-18: Signal and Job Control
Parent Process PID: 1932
Child process created. PID: 1933
Child Process Started. PID: 1933
Child process is running...
Child process is running...

Sending SIGSTOP to child...
Child process stopped successfully.

Sending SIGCONT to resume child...
Continue signal sent successfully.
Child process is running...

Sending SIGTERM to terminate child...
Child terminated by signal: 15
Parent process completed.
```

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ nano skill18_job_control.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ sleep 100 &
[1] 1941
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ kill -STOP $!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ jobs -l
[1]+ 1941 Stopped (signal) sleep 100
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ kill -CONT $!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ jobs -l
[1]+ 1941 Running sleep 100 &
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ kill $!
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ jobs
[1]+ Terminated sleep 100
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-18$ |
```